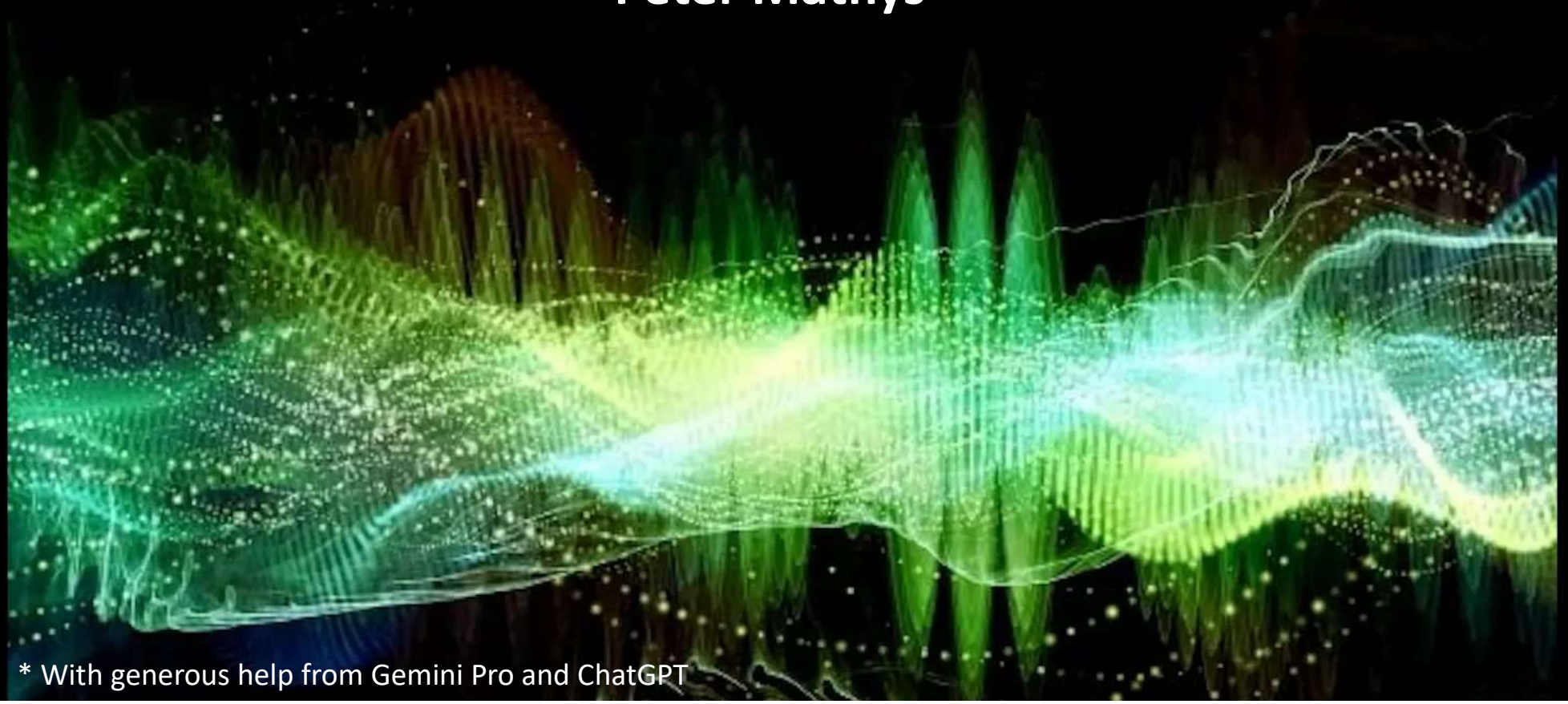
ECEN 4/5632 Theory and Application of Digital Filters
Lecture 25, 11-10-26

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* With generous help from Gemini Pro and ChatGPT

Feature Engineering

From Physics to Perception

STFT → Mel filterbanks → log compression → MFCCs → fixed-size ML inputs

WEEK 13 • LECTURE 25 • 75 MINUTES

Week 13 changes the objective

THE TRANSITION FROM DSP ANALYSIS TO MACHINE LEARNING

00–15 MIN

- Classical DSP asks: can we represent, filter, or reconstruct the signal faithfully?
- ML asks: which representation makes the decision boundary easiest to learn?
- A feature may intentionally discard information that is irrelevant to the task.
- But discarded information cannot be recovered later—feature engineering encodes assumptions.

Preserve what matters for the label; compress what does not.

Feature engineering = physics + perception + task knowledge

Instructor cue: Ask for one task where perfect reconstruction is unnecessary: siren detection, keyword spotting, bearing-fault classification.

Learning outcomes

BY THE END OF LECTURE 21

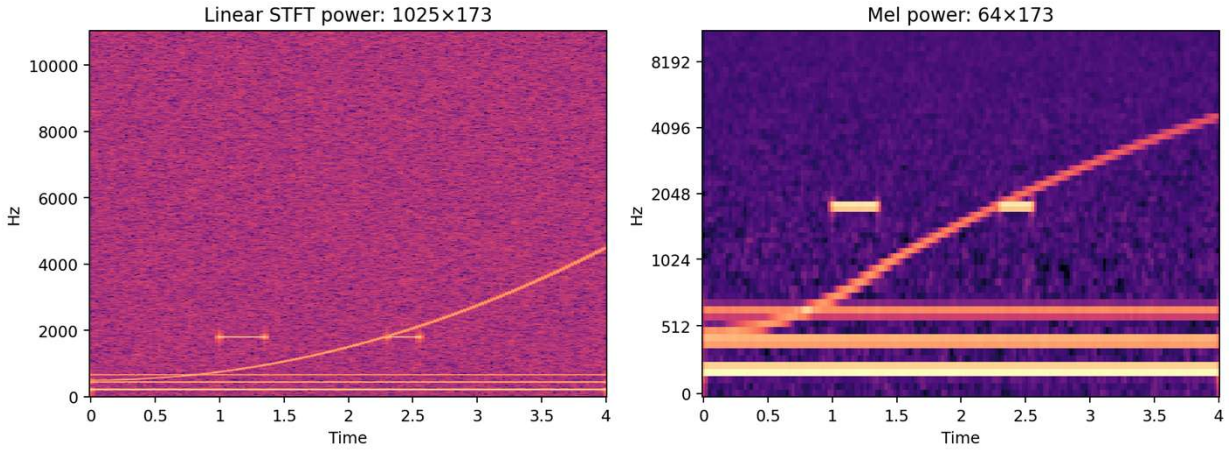
00–15 MIN

- Explain why linear-Hz STFT bins are not always the most efficient coordinates for audio classification.
- Construct a Mel-spectrogram conceptually as a weighted sum of STFT power bins.
- Explain log compression and the DCT step used to form MFCCs.
- Standardize waveform geometry: channels, sample rate, duration, and feature shape.
- Choose between Mel-spectrograms, MFCC matrices, and summarized MFCC vectors for different model families.

The problem with linear frequency

THE STFT IS PHYSICALLY NATURAL, NOT PERCEPTUALLY UNIFORM

00–15 MIN



- At 22.05 kHz with n_fft=2048, the one-sided STFT has 1025 frequency bins.
- For many audio tasks, fine resolution at the top of the band is less useful than fine resolution at low frequencies.
- Mel filterbanks pool neighboring bins nonuniformly, reducing dimension while keeping time structure.

$$\text{STFT power } P[k,m] \rightarrow \text{Mel power } S_{\text{mel}}[b,m]$$

Instructor cue: Phrase "human hearing is logarithmic" carefully: Mel is a perceptual warping, not a universal law of hearing.

Physics coordinates vs perception-oriented coordinates

SAME SOUND, DIFFERENT REPRESENTATIONS

00–15 MIN

Linear Hz / STFT
Equal-width frequency bins
Direct physical meaning
Filter-analysis friendly

Mel
More low-frequency detail
Broader high-frequency bands
Compact audio front end

MFCC
DCT of log-Mel
Compact spectral envelope
Classical baseline friendly

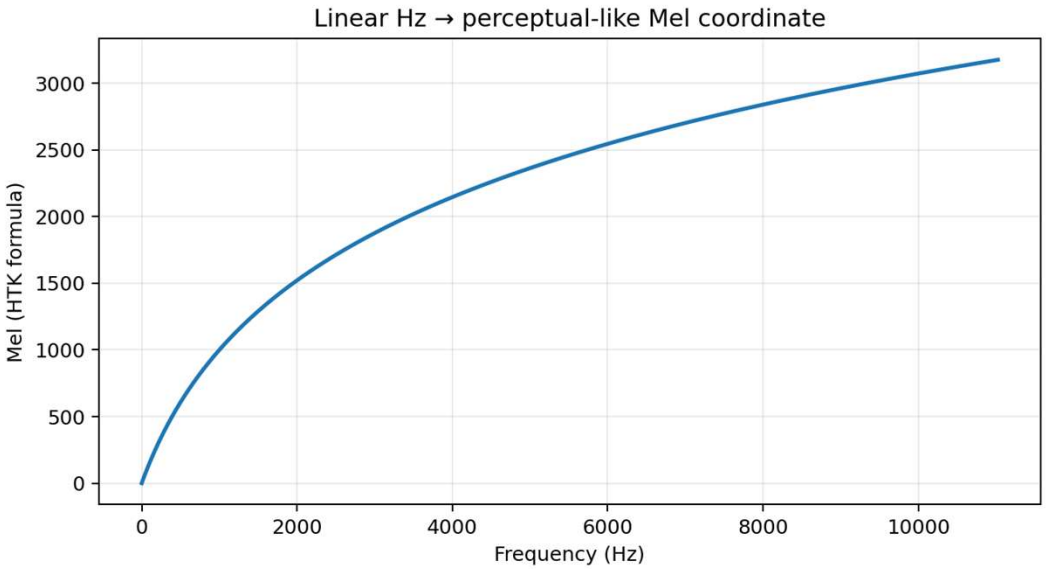
CNN input
Keep full 2D map
Learn local patterns
Next week: CNNs

No representation is free: every compression trades detail for invariance or efficiency.

The Mel scale

ONE COMMON MAPPING FROM HZ TO A PERCEPTUAL-LIKE COORDINATE

15–35 MIN



$$m = 2595 \log10(1 + f/700)$$

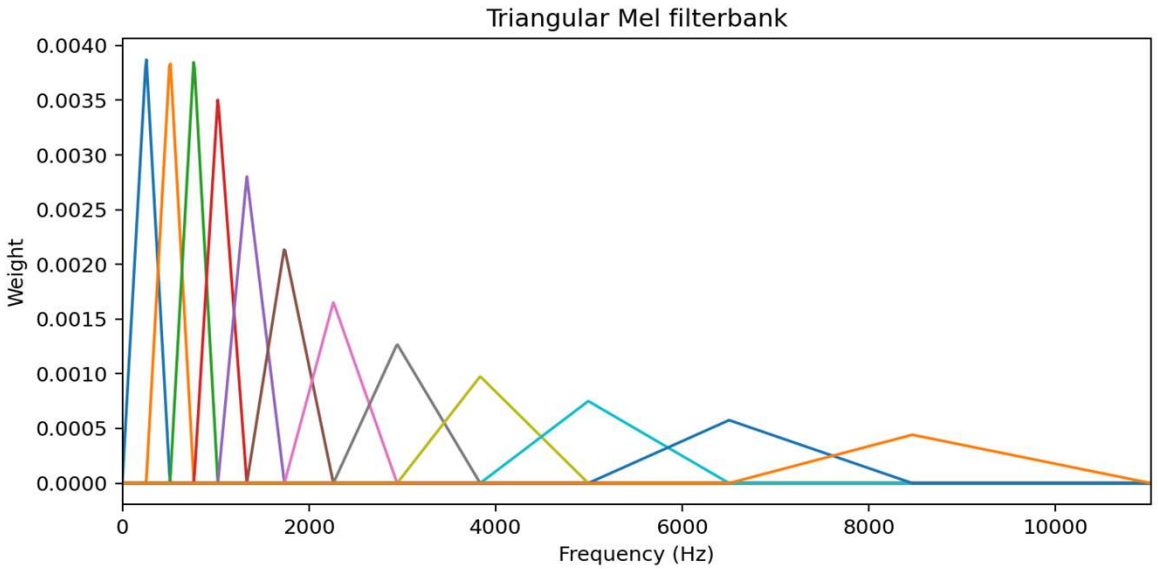
- The formula shown is the common HTK-style mapping.
- librosa defaults to a Slaney-style Mel definition; set htk=True for the HTK convention.
- The important idea is frequency warping—not memorizing one constant.

Instructor cue: Use 100→200 Hz versus 10,000→10,100 Hz as intuition; avoid claiming an exact perceptual threshold.

Mel filterbanks are weighted pooling

TRIANGULAR BANDS COMPRESS FREQUENCY WITHOUT DESTROYING TIME

15–35 MIN



- Each filter H_b[k] assigns weights to neighboring FFT bins.
- Low-frequency filters are relatively narrow; high-frequency filters cover more Hz.
- Overlapping filters provide smooth transitions between bands.

$$S_mel[b,m] = \Sigma_k \, H_b[k] \, |X[m,k]|^2$$

Matrix view: Mel features are a linear map of power

A BRIDGE BACK TO LINEAR ALGEBRA

15–35 MIN



- Frequency compression happens independently in every time frame.
- Time resolution is inherited from the STFT hop length.
- Choosing n_mels controls the frequency-feature dimension.

librosa: compute a Mel-spectrogram

CODE SHOULD MIRROR THE MATH

15–35 MIN

```
M = librosa.feature.melspectrogram(
            y=y, sr=22050,
      n_fft=2048, hop_length=512,
              n_mels=64,
              power=2.0,
                 )
M_db = librosa.power_to_db(M, ref=np.max)

  print(M.shape)      # (64, time_frames)
print(M_db.max())   # 0 dB relative to peak
```

Instructor cue: Ask students to predict the first dimension. power=2 means power; power_to_db therefore uses 10 log10.

Why take the logarithm?

DYNAMIC RANGE COMPRESSION IS PART OF FEATURE DESIGN

15–35 MIN

$$S_{dB} = 10 \log_{10}(S / S_{ref})$$

- Spectral power can span orders of magnitude.
- Log compression turns ratios into differences and makes weak structure more visible.
- For power quantities use $10 \log_{10}(\cdot)$; for amplitude ratios use $20 \log_{10}(\cdot)$.
- A floor is required numerically because log(0) is undefined.

100× power ratio → +20 dB

0.01× power ratio → −20 dB

MFCC pipeline

A COMPACT DESCRIPTION OF SPECTRAL-ENVELOPE SHAPE

35–55 MIN



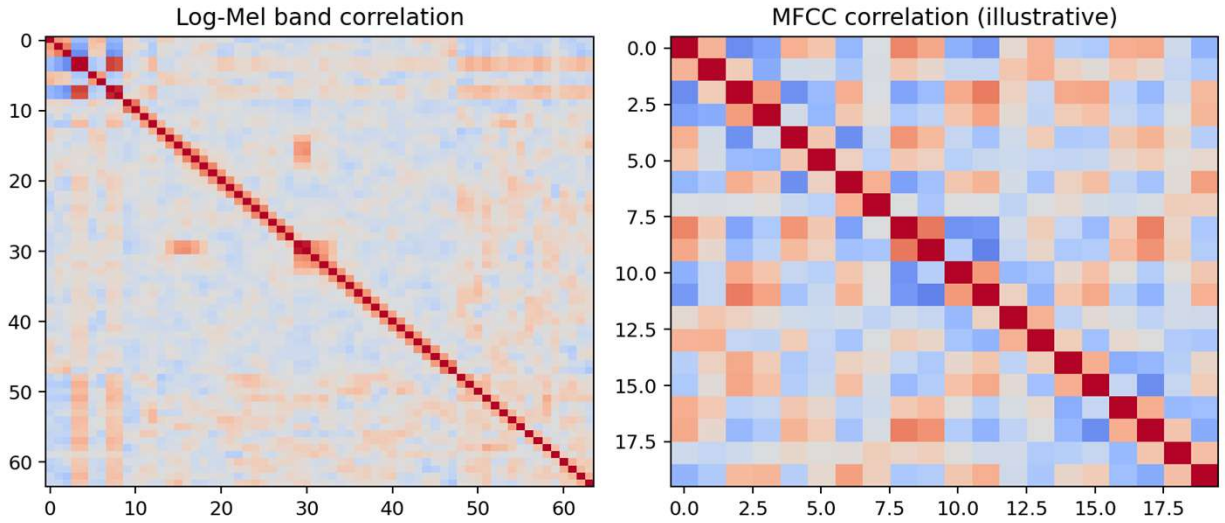
$$C[n,m] = \mathrm{DCT}_b\{ \log S_{\mathrm{mel}}[b,m] \}$$

- Many classical pipelines retain roughly the first 13–40 coefficients.
- Coefficient 0 largely tracks overall log-energy; higher indices encode faster variation across Mel bands.

What does the DCT buy us?

APPROXIMATE DECORRELATION AND ENERGY COMPACTION

35–55 MIN



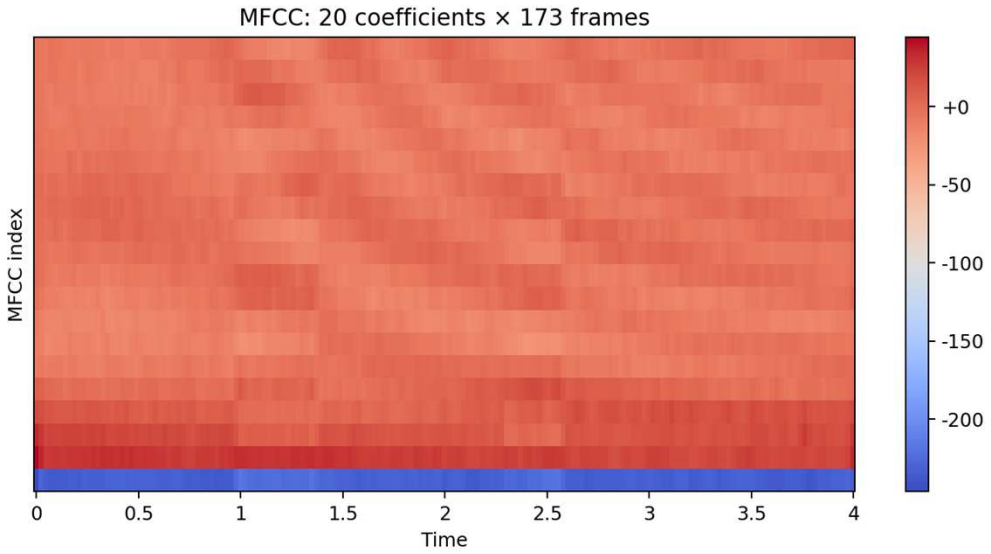
- Neighboring Mel bands overlap, so their log energies are correlated.
- The DCT rotates those coordinates into cosine-basis directions.
- Low-order coefficients often concentrate useful smooth spectral-envelope information.
- “Decorrelates” is approximate and data-dependent—not a universal guarantee.

Instructor cue: Connect to older Gaussian/diagonal-covariance models. The DCT is not magic statistical independence.

How should students interpret MFCC indices?

USE QUALITATIVE INTUITION, NOT ONE-TO-ONE PHYSICAL LABELS

35–55 MIN



- Low-order coefficients: broad spectral-envelope / tilt information.
- Higher-order coefficients: progressively finer variation across Mel bands.
- MFCCs intentionally discard much detailed harmonic structure.
- That compression can help a classical classifier—or hurt a task needing fine time–frequency detail.

Mel-spectrogram or MFCC?

CHOOSE THE REPRESENTATION TO MATCH THE MODEL AND TASK

35–55 MIN

Mel-spectrogram
2D time–frequency geometry
Natural CNN input
More local detail

MFCC matrix
Compact spectral-envelope view
Still has time frames
Classical baseline friendly

MFCC summary
Mean/std over time
Fixed vector
Temporal order destroyed

No universally “best” feature: representation creates an inductive bias.

Standardizing waveform geometry

BEFORE FEATURES, MAKE EXAMPLES COMPARABLE

55–75 MIN

1. **Channels**
   Stereo → mono or a defined multi-channel policy
2. **Sample rate**
   Resample every clip to 22.05 kHz
3. **Duration**
   Pad/truncate to exactly 4.0 s
4. **Amplitude**
   Use a consistent numeric range; do not erase meaningful level without justification
5. **Features**
   Same n_fft, hop, n_mels, and n_mfcc for every clip

Downmixing stereo to mono

A CHANNEL DECISION IS PART OF THE MEASUREMENT MODEL

55–75 MIN

$$y_mono[n] = 0.5(y_L[n] + y_R[n])$$

- Simple averaging is common when spatial information is irrelevant.
- Do not downmix blindly if direction or spatial cues are part of the label.
- librosa.load(..., mono=True) can downmix automatically.

Feature engineering begins with deciding what information you are willing to throw away.

Resampling: one physical frequency grid for every example

22.05 KHZ BECOMES THE COURSE CONVENTION

55–75 MIN

$$N_samples = duration \times f_s = 4.0 \times 22050 = 88{,}200$$

- A fixed sample rate makes FFT-bin frequencies and Mel bands comparable across files.
- Always use an anti-aliased resampler; naive sample dropping is not resampling.
- 22.05 kHz implies Nyquist = 11.025 kHz—acceptable only if the task does not need higher-frequency information.

DSP prerequisite
Week 2 still matters: bad rate conversion can create aliases.

Padding and truncation: force a fixed tensor shape

SAME DURATION → SAME NUMBER OF FRAMES

55–75 MIN

```
target_len = int(4.0 * 22050)   # 88,200 samples

            if len(y) < target_len:
        y = np.pad(y, (0, target_len - len(y)))
                    else:
                y = y[:target_len]

          assert y.shape == (target_len,)
```

- Padding location is a design choice: right, left, or centered.
- Truncation can remove events—random crops or multiple crops may be better during training.
- For this week, deterministic right-padding/right-truncation keeps the baseline easy to debug.

Shape bookkeeping: predict dimensions before running code

A POWERFUL DEBUGGING HABIT

55–75 MIN



- Exact frame count depends on STFT centering/padding conventions.
- Flattening 20×173 MFCCs gives 3460 features; mean+std gives only 40.
- Dimensionality reduction is useful—but every reduction throws away structure.

Live code-along: one UrbanSound8K clip

MAKE EVERY PREPROCESSING DECISION VISIBLE

55–75 MIN

```
y, sr0 = librosa.load(path, sr=None, mono=True)
y = librosa.resample(y, orig_sr=sr0, target_sr=22050)
y = fix_length(y, int(4.0 * 22050))

mel = librosa.feature.melspectrogram(
y=y, sr=22050, n_fft=2048,
hop_length=512, n_mels=64, power=2.0)
mel_db = librosa.power_to_db(mel, ref=np.max)
mfcc = librosa.feature.mfcc(S=mel_db, n_mfcc=20)
```

Instructor cue: Pause after every line: what invariant did we establish, and what information might we have discarded?

Concept check

REASON ABOUT REPRESENTATION CHOICES

CHECK

- A. Increasing n_mels always improves classification accuracy.
- B. Converting stereo to mono is harmless for every audio task.
- C. Mean MFCCs reduce dimensionality but remove temporal ordering.
- D. Converting power to dB restores information lost by Mel pooling.

Lecture 25 synthesis

FROM WAVEFORM PHYSICS TO ML-READY FEATURES



- Mel features compress frequency according to a perceptual warping.
- MFCCs further rotate/compress log-Mel information for compact classical features.
- Fixed channels, rate, duration, and feature parameters give the model consistent geometry.
- Next lecture: feature matrices become X, labels become y, and evaluation discipline becomes central.